6.13 Evaluating Exponents

Lesson Introduction

Benchmark	MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.	Learning Targets	• I can evaluate integers with natural number exponents. • I can evaluate a positive rational number raised to a power.

Benchmark Coherence	Building On	Working Toward
	N/A	MA.7.NSO.1.1 Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Essential Questions	• How do you represent repeated multiplication using exponents? • How can you represent exponential expressions as repeated multiplication? • How do you determine whether an exponential expression will have a positive or negative value?
Lesson Narrative	In this lesson, students extend and apply their knowledge of multiplying integers and of exponential expressions to evaluate exponential expressions with integer bases. Students explore how to determine whether an exponential expression will have a positive or negative value and practice rewriting exponential expressions, including those with positive decimal and fractional bases, as repeated multiplication and vice versa.

Component Summary				
	6.13.1 Warm-Up (~5 min.)	Describe what they want to get better at after watching a motivational video *(SE p. 291)*	6.13.3 Bring It Together (10 min.)	Discover when the signs of exponential expressions will be positive and negative *(SE p. 293)*
	6.13.2 Exploration (10 min.)	Explore how to evaluate exponential expressions with positive and negative bases *(SE p. 292)*	6.13.4 Your Turn! (10 min.)	Additional practice raising rational numbers to powers *(SE p. 294)*
	6.13.5 Wrap-Up (~5 min.)	Self-assessment rating of levels of understanding of learning targets from the unit *(SE p. 294)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Base • Exponent/Power • Factors • Value	• (Optional) Chart paper • (Optional) Markers • (Optional) Note cards	• Warm-Up 6.13.1 contains a video to accompany the question prompts. • Consider creating an anchor chart with instructions, examples, and visual models.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not correctly evaluate exponential expressions when there is a negative sign inside the parentheses. For example, $(-4)^2 \neq -16$.
Students may think that an exponent of 1 means to multiply the number by itself once. For example, $16^1 = 16 \times 16$.</td><td>Consider displaying and adding to the anchor chart created with Lesson 1 with instructions, examples, and visual models of evaluating exponential expression in 6.13.2 Exploration.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
In 6.13.3 Bring It Together, have students discuss their work and answers after first working individually. The discussions will help students gain confidence and understanding.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
6.13.4 Guided Practice question 3 provides an opportunity for students to analyze errors in samples of student work. Students identify and correct the errors. Students will be less likely to make the mistake themselves after completing this analysis.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">6.13.4 Your Turn! includes fractions raised to powers for students to evaluate. Notice benchmark MA.6.NSO.3.3 includes “rational numbers” in the benchmark language, as fluency with fractions and decimals is an expectation by the end of Grade 6. This is spiraled from Lesson 1 in this unit.

Consider utilizing the On-Ramp to 6th Grade learning tool to provide scaffolding for students in need of additional support when evaluating expressions.
Evaluate Expressions</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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6.13.1 Warm-Up: Consider the YET

Component Overview: *Play motivational video "The Power of Yet" for students. Then have students reflect by answering the following questions.*

6.13.1 Warm-Up: Consider the YET

1. Consider the belief that "you can get better at whatever you want to get better at." Explain whether you agree or disagree.
   Answers will vary.

2. Consider the quote, "I've never heard of someone who took up running and didn't get faster the more they did it." That person just may not have been a fast runner ... YET.

   Complete the statement to describe something you want to get better at.
   Answers will vary.
   I ____

   ____...YET!

Consider providing an auditory entry point by having students discuss question 1. Write sentence stems on the board to encourage conversation, such as "I agree with the statement because…" or "I disagree with the statement because…."

Prompt students to reference a specific moment in this unit in which they exhibited growth mindset or heard their partner or classmate exhibit growth mindset.

6.13.2 Exploration: Building on the Understanding of Exponents

Component Overview: *Review the vocabulary as a class before students complete questions 1 and 2 independently. Students then work with a partner to explore how to evaluate exponential expressions with positive and negative bases.*

6.13.2 Exploration: Building on the Understanding of Exponents

Recall the vocabulary of exponential expressions.

value (of the expression)

exponent (or power)

$$3^4 = \underbrace{3 \cdot 3 \cdot 3 \cdot 3}_{\text{factors}} = 81$$

base

1. Think of a number that, when raised to a power less than 6, will result in a value of 16.

2. Write the base and its exponent.
   Answers will vary.

   $$\boxed{2}^{\boxed{4}}$$

   Answers could also include $(-2)^4$, $(4)^2$, *or* $(-4)^2$.

3. Compare your expression with your partner and check for accuracy.

4. If your partner found a different base and exponent that meets the description, write it below. If your partner wrote the same base and exponent as you, work together to find a second exponential expression that also works.
   Answers will vary. See examples from question 2.

   $$\boxed{}^{\boxed{}}$$

Turn and Talk
Prompt students to discuss vocabulary terms, their expressions, and possible values.

What are some factors of 16 that you could use as a base? Is there more than one exponent and base combination that will equal 16? How do you know?

Remind students who may be struggling how to find the prime factorization of a number. Suggest using the prime factorization of 16 to determine possible bases and exponents to use.

6.13.2 Exploration: Building on the Understanding of Exponents (continued)

5. With your partner, choose one of the exponential expressions used and rewrite it making the base negative.
   Answers will vary.

$$(-4)^{\boxed{2}}$$

6. Explain whether the negative base changes the value of the expression.
   Answers will vary. The exponent is even, so the value is the same whether the base is negative or positive. The negative base does not change the value of the expression because when you multiply an even number of negative factors, the sign of the product is positive. Therefore, the value is still 16.

7. Discuss with your partner how you could change the exponent to result in a negative answer. Write a brief summary of your discussion below.
   Answers will vary. You can change the exponent to an odd number to result in a negative answer because when you multiply an odd number of negative factors, the product will be negative.

How does a negative base change the value of the expression when the exponent is even? How does it change the value when the exponent is odd? Why do you think that is? (See extension in question 6 for 6.13.3.)

If students struggle expressing themselves in writing, consider allowing students to respond to the questions verbally first, taking turns with a partner. Students can then respond to questions 6 and 7 as a synthesis rather than a starting point.

The next component will solidify understanding for questions 6 and 7, so it is not necessary that students are able to answer these accurately at this point. The goal is for them to discuss their thinking related to what they have learned already in the unit. Listen as students discuss to bring their thinking into the next component.

6.13.3 Bring It Together: Building on the Understanding of Exponents

Component Overview: *Students build their understanding of evaluating exponents and how the rules of multiplication with integers apply when the signs of the bases are positive and negative.*

6.13.3 Bring It Together: Building on the Understanding of Exponents

When evaluating exponents with a negative base, the same rules of multiplication with integers apply.

1. In the expression $(-4)^2$, the base is -4 and the exponent is 2, resulting in the repeated multiplication sentence of $(-4)(-4)$.

2. Why do the results of $(-4)^2$ result in a positive number?
   Answers will vary. The product of two negative integers is positive.

3. In the expression $(-4)^3$, the base is -4 and the exponent is 3, resulting in the repeated multiplication sentence of $(-4)(-4)(-4)$.

4. Why do the results of $(-4)^3$ result in a negative number?
   A positive times a positive = a positive.
   $$(-4)(-4) = 16$$
   Then, a positive times a negative = a negative.
   $$(16)(-4) = -64$$

Utilize inquiry to position students for synthesis.
- *What impact does the negative sign have on the value of the outcome?*
- *How does that change if the negative sign was the coefficient on the outside of the parenthesis? Provide an example.*

If students struggle expressing themselves in writing, consider providing sentence starters for question 4, such as "A negative times a negative equals a ______," "A positive times a negative equals a ________," or "Therefore, the answer must result in an _______ answer."

6.13.3 Bring It Together: Building on the Understanding of Exponents (continued)

5. For each expression, determine whether its value is positive or negative.

Expression	Positive or Negative Value?
$(-2)^2$	*Positive*
$(-2)^3$	*Negative*
$(-2)^4$	*Positive*
$(-2)^5$	*Negative*
$(-2)^{19}$	*Negative*

Notice and Wonder
To encourage the connection between the value of the exponent and sign of the base, consider asking students what they notice and wonder about the sign of the values of each exponential expression.

6. When evaluating exponents with a negative base, the value of the expression will be [○ positive / ○ negative] if the exponent is an [○ odd / ○ even] number, and the value of the expression will be [○ positive / ○ negative] if the exponent is an [○ odd / ○ even] number.

Note there are two correct answers for question 6. If students complete the first blank with "negative," then the remaining blanks would be "odd," "positive," and "even," respectively.

6.13.4 Your Turn!

Component Overview: *Students independently evaluate exponential expressions and rewrite repeated multiplication expressions in exponential form. Students also assess whether two expressions are equivalent and show their work to justify their responses.*

6.13.4 Your Turn!

1. Evaluate each expression.

a. $(-3)^4$ — 81

b. $\left(\frac{15}{17}\right)^2$ — $\frac{225}{289}$

c. $(-1)^5$ — -1

2. Rewrite each repeated multiplication sentence as an exponential expression. Then, find the value of the expression.

a. $(-12)(-12)(-12)$

$(-12)^3$

$= -1{,}728$

b. 3.4×3.4

$(3.4)^2$

$= 11.56$

3. Determine if each equation in the table is true or false. Show your work to support your answer.

Equation	True or False?	Work
$16^1 = (-8)^2$	False	$16^1 = 16$ $(-8)^2 = (-8)(-8)$ $= 64$
$(-5)^3 = -5 \cdot 5 \cdot 5$	True	$(-5)^3 = -5 \cdot -5 \cdot -5$ $-5 \cdot -5 \cdot -5 = 25 \cdot -5$ $= -125$ $-5 \cdot 5 \cdot 5 = -25 \cdot 5$ $= -125$

Consider displaying the anchor chart created in Lesson 1 with instructions, examples, and visual models for students to reference.

Alternatively, offer students a station to opt-in for support using On-Ramp to 6th Grade videos to learn alongside Study Experts.

Prompt students to use the word "equivalent" in their verbal explanations of their work or findings.

6.13.5 Wrap-Up: Reflection

Component Overview: *Students complete a self-assessment, rating their level of understanding of the topics covered throughout the unit.*

6.13.5 Wrap-Up: Reflection

1. In this unit, you have learned multiple concepts about operations with integers. Rate yourself on the following skills based on your level of understanding.
   Answers will vary.

Statement	**Rating** 1 – I struggle with this. 2 – I have some skills. 3 – I am almost there. 4 – I have got this!
I can evaluate positive rational numbers and integers raised to a power.	
I can express composite whole numbers as a product of prime factors.	
I can add integers.	
I can subtract integers.	
I can multiply integers.	
I can divide integers.	

Consider having students create and solve examples of each learning target they rate as a 4 or 5. Ask students to consider a plan for filling in gaps in their learning for targets they rated as a 1, 2, or 3.